

1. (8 marks) Determine whether the set of functions $\{x^2 + 2x + 3, -x^2 + 2x, 4x + 9\}$ in P_2 are linearly independent. Do NOT use the Wronskian. NOTE: You must show all your calculations, and you MUST show each step of how the resulting matrix was set up. If you do this question only by inspection (i.e. trial and error or guessing), or do not show all your work, you will NOT receive marks.

SOLN: Consider $k_1 \vec{p}_1 + k_2 \vec{p}_2 + k_3 \vec{p}_3 = \vec{0}$

$$k_1(x^2 + 2x + 3) + k_2(-x^2 + 2x) + k_3(4x + 9) = \vec{0}$$

$$\Leftrightarrow (k_1 - k_2)x^2 + (2k_1 + 2k_2 + 4k_3)x + 3k_1 + 9k_3 = \vec{0}$$

Equate
coeff.

$$x^2: k_1 - k_2 = 0$$

$$x^1: 2k_1 + 2k_2 + 4k_3 = 0$$

$$x^0: 3k_1 + 9k_3 = 0$$

*
show
this

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 2 & 4 \\ 3 & 0 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

A square

$$\det(A) = \begin{vmatrix} 1 & -1 & 0 \\ 2 & 2 & 4 \\ 3 & 0 & 9 \end{vmatrix} \begin{vmatrix} 1 & -1 \\ 2 & 2 \\ 3 & 0 \end{vmatrix}$$

$$= 18 - 12 + 0 - 0 - 0 + 18$$

$$= 24$$

$$\neq 0$$

$\therefore A$ is invertible

$$\therefore k_1 \vec{p}_1 + k_2 \vec{p}_2 + k_3 \vec{p}_3 = \vec{0} \quad ("A\vec{x} = \vec{0}")$$

has only the trivial soln.

$\therefore \vec{p}_1, \vec{p}_2, \vec{p}_3$ are linearly independent

OR could do row reduction

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 0 & 0 \\ 3 & 0 & 9 & 0 & 0 & 0 \end{array} \right] \xrightarrow[R_3: R_3 - 3R_1]{R_2: R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 4 & 4 & 0 & 0 & 0 \\ 0 & 3 & 9 & 0 & 0 & 0 \end{array} \right] \xrightarrow[R_2: R_2 - 3R_3]{R_2: \div 4} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 3 & 9 & 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 & 0 \end{array} \right]$$

$6k_3 = 0 \Rightarrow k_3 = 0$
 $k_2 + k_3 = 0 \Rightarrow k_2 = 0$
 $k_1 - k_2 = 0 \Rightarrow k_1 = k_2 = 0$
 $\therefore$ trivial soln only $\therefore$ lin indep

2. (6 marks) You are given that the reduced row-echelon form of

$$A = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 \\ 3 & -7 & 2 & 0 & 1 \\ 2 & -5 & 2 & 4 & 6 \\ 4 & -9 & 2 & -4 & -4 \end{bmatrix} \text{ is } \begin{bmatrix} 1 & 0 & -4 & -28 & -37 \\ 0 & 1 & -2 & -12 & -16 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Find the column space basis, row space basis, and nullspace basis of A . You must show all your work for obtaining the nullspace basis.

for nullspace basis consider

$$\left[\begin{array}{ccccc|c} 1 & 0 & -4 & -28 & -37 & 0 \\ 0 & 1 & -2 & -12 & -16 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Let $x_5 = t \in \mathbb{R}$

$x_4 = s \in \mathbb{R}$

$x_3 = r \in \mathbb{R}$

$x_2 = 2r + 12s + 16t$

$x_1 = 4r + 28s + 37t$

So $\vec{x} = \begin{bmatrix} 4 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} 28 \\ 12 \\ 0 \\ 0 \\ 0 \end{bmatrix} s + \begin{bmatrix} 37 \\ 16 \\ 0 \\ 0 \\ 0 \end{bmatrix} t$

Final Answers: Column Space Basis:

$$\begin{bmatrix} -1 \\ 3 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ -7 \\ -5 \\ -9 \end{bmatrix}$$

Row Space Basis: $[1 \ 0 \ -4 \ -28 \ -37], [0 \ 1 \ -2 \ -12 \ -16]$

Nullspace Basis: $\begin{bmatrix} 4 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 28 \\ 12 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 37 \\ 16 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

3. (6 marks; 3 marks each) Answer each question in the space provided.

- a) Consider the set $F(-\infty, \infty)$ of all real-valued functions that are defined on $(-\infty, \infty)$. Suppose that scalar multiplication for this set is defined as usual, but that addition is defined instead by $(f + g)(x) = f(x) + 2g(x)$. **Prove** that the following vector space property does **NOT** hold: " $((f+g)+h)(x) = ((f+g)+h)(x)$ "

SOLN:

Consider $\vec{f} = \vec{g} = \vec{h} = 1 \in F$

LHS = $(\vec{f} + (\vec{g} + \vec{h}))(x)$
 $= g(x) + 2h(x)$
 $= f(x) + 2(g(x) + 2h(x)) = 7$

optional to do counter-ex. could prove this in general

RHS = $((\vec{f} + \vec{g}) + \vec{h})(x)$
 $= f(x) + 2g(x)$
 $= f(x) + 2g(x) + 2h(x) = 5$

$\therefore LHS \neq RHS$
 So this property fails

- b) Suppose that the set W consists of all 3×3 matrices of the form $\begin{bmatrix} b & 0 & b \\ 0 & b & 0 \\ b & 0 & b \end{bmatrix}$ where

b may be any scalar. Suppose that scalar multiplication and addition for this set is defined as usual. Determine whether this set is closed under scalar multiplication (i.e. If $\mathbf{u}$ is an object in W and k any scalar, then $k\mathbf{u}$ is in W) and **prove** your answer.

SOLN: Let $A = \begin{bmatrix} b & 0 & b \\ 0 & b & 0 \\ b & 0 & b \end{bmatrix} \in W$, let $k \in \mathbb{R}$.

Consider $kA = \begin{bmatrix} kb & 0 & kb \\ 0 & kb & 0 \\ kb & 0 & kb \end{bmatrix} \in W$ ✓

$\therefore W$ is closed under scalar multiplication

still of correct form above. 3×3 with 0's in desired entries

4. (8 marks total) Answer each question in the space provided. Show ALL your work.

- a) (2 marks) The planes $-9(x-1) + 21(y+3) - 12(z-4) = 0$ and $3x - 7y + 4z = 8$ are parallel. Answer TRUE or FALSE, and justify your answer. You will receive credit ONLY if your answer is correct AND properly justified.

SOLN:

Answer: TRUE

Justification: normal to plane ① is $\vec{n}_1 = (-9, 21, -12)$
normal to plane ② is $\vec{n}_2 = (3, -7, 4)$

Notice that $\vec{n}_1 = -3\vec{n}_2$; they are scalar multiples of one another

- b) (3 marks) Use the Wronskian to determine whether the functions $2x$ and $5x^3 + 7$ are linearly independent. Other methods will NOT receive marks.

Hint: In general, $W(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_n(x) \\ f_1'(x) & f_2'(x) & \dots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{vmatrix}$

SOLN:

$$W(x) = \begin{vmatrix} 2x & 5x^3 + 7 \\ 2 & 15x^2 \end{vmatrix}$$

$$= 30x^3 - 10x^3 - 14$$

$$= 20x^3 - 14$$

Since $W \neq 0$ (at least not for all x),
 $\therefore$ lin indep.

- c) (2 marks) Under what conditions on scalars b and c are the vectors $(5b, -1, 2)$ and $(0, 3c, 7)$ orthogonal? If there are no restrictions, write N/A.

SOLN: Consider $(5b, -1, 2) \cdot (0, 3c, 7) = 0 - 3c + 14$

"orthogonal" $\Leftrightarrow 0 - 3c + 14 = 0$
 $\Leftrightarrow c = \frac{14}{3}$

$\therefore$ $\boxed{\begin{matrix} c = 14/3 \\ b \text{ N/A} \end{matrix}}$

- d) (1 mark) Find the distance between the points $(1, 3)$ and $(-5, 2)$.

SOLN:

$$d = \sqrt{(1+5)^2 + (3-2)^2}$$

$$= \sqrt{6^2 + 1^2}$$

$$= \sqrt{37}$$

5. (12 marks total) Answer each of the following in the space provided. You do NOT have to show your work; only the final answer will be marked.

- a) (3 marks)** Find the parametric equations of the line through the point $(3, 1, 2)$ and perpendicular to the plane $6(x-3) + 4(y-5) + 6z = 0$.

Answer: $x = 3 + 6t$ $y = 1 + 4t$ $z = 2 + 6t$, where $t \in \mathbb{R}$

- b) (1 mark)** Find the value(s) of x such that $(-1, x, 6)$ has length 10.

Answer: $x = \pm \sqrt{63} = \pm 3\sqrt{7}$

- c) (2 marks)** Find the cross product of $\mathbf{u} = (a, 5, 1)$ and $\mathbf{v} = (-1, 3, 4)$ where a is a constant.

Answer: $\mathbf{u} \times \mathbf{v} = (17, -1-4a, 3a+5)$

- d) (2 marks)** Find a basis for the set of all 2×2 matrices of the form

$$\begin{bmatrix} a & a+b \\ -3b & 0 \end{bmatrix} \text{ where } a, b \text{ are real numbers.}$$

Answer: $\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ -3 & 0 \end{bmatrix} \right\}$

- e) (2 marks)** Suppose A is a 12×4 matrix. What is the maximum rank that A could have? What is the minimum nullity that A^T could have?

Answers: $\max \text{rank}(A) = 4$, $\min \text{nullity}(A^T) = 8$.

- f) (1 mark)** Consider the set V consisting of all 2×2 lower triangular matrices. Find the dimension of V .

Answer: $\dim(V) = 3$

- g) (1 mark)** Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Find the rank of A^2 .

Answer: $\text{rank}(A^2) = 1$